

Program 7

Step 1:- Start Minikube and check its Status

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...

🔥 Docker is nearly out of disk space, which may cause deployments to fail! (89% of capacity). You can pass '--force' to skip this check.
💡 Suggestion:

Try one or more of the following to free up space on the device:

1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
   Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
🔗 Related issue: https://github.com/kubernetes/minikube/issues/9024

❗ Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
🌞 Enabled addons: default-storageclass, storage-provisioner, metrics-server
🎉 Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions$ minikube status
minikube
type: Control Plane
host: Running
kubernetes: Running
apiserver: Running
kubeconfig: Configured
```

Step 2:- Create nginx-deployment.yaml

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ gedit nginx-deployment.yaml
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ ls
nginx-deployment.yaml
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 3:- Apply nginx-deployment

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

Step 4:- Check Status

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2             2           11s
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-bprbr   1/1     Running   0           16s
nginx-deployment-6f9664446b-bvhgj   1/1     Running   0           16s
```

Step 5:- Expose Deployment as a service

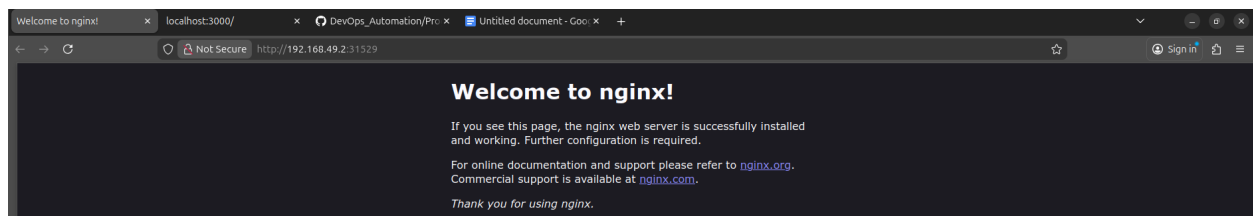
```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl expose deployment nginx-deployment \
  --type=NodePort \
  --port=80
service/nginx-deployment exposed
service/nginx-deployment exposed
bash: service/nginx-deployment: No such file or directory
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes           ClusterIP   10.96.0.1     <none>        443/TCP          144m
nginx-deployment     NodePort    10.106.28.141 <none>        80:31529/TCP     26s
```

Step 6:- Access the Webapp

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31529

 Opening service default/nginx-deployment in default browser...



Step 7:- Delete all pods and Deployments

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-bprbr" deleted from default namespace
pod "nginx-deployment-6f9664446b-bvhgj" deleted from default namespace
pod "program-8" deleted from default namespace
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl delete svc --all
service "kubernetes" deleted from default namespace
service "nginx-deployment" deleted from default namespace
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog7$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted from default namespace
```